

Aaron Lin

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EDUCATION

Boston University

B.S. Mechanical Engineering, Aerospace Concentration

GPA: 3.76/4.0 | Dean's List

Boston, MA

Sep 2024 – May 2028

Relevant Coursework: Mechanics of Materials, Energy and Thermodynamics, Differential Equations, Computational Linear Algebra
In Progress (Fall 2026): Fluid Mechanics, Engineering Mechanics 2 (Dynamics), Thermodynamics & Statistical Mechanics, Electromechanical Design, Manufacturing Processes

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern

Feb 2026 – Jun 2026

Culturon

Sydney, Australia

- Executed Optical Emission Spectroscopy (OES) experiments using an HR4000 spectrometer and authored technical protocols to evaluate industrial plasma consistency and detect contaminants.
- Developed an automated Python analysis suite (pandas, scipy, astroquery) to process OES spectral data, identify elemental peaks, and cross-reference findings with the NIST atomic database in real-time.
- Authored a technical report benchmarking OES plasma diagnostics against an existing reference chamber system, and identifying key emission lines for tracking contamination and process consistency going forward.

AI Trainer

Oct 2025 – Dec 2025

Handshake AI Fellowship

Remote

- Developed and evaluated domain-specific prompts to benchmark LLM accuracy across scientific and engineering subfields.
- Contributed to improving AI understanding of complex biological topics through expert review and feedback.
- Conducted independent research to support prompt development and evaluation tasks.

PROJECTS

Project Engineer

Sep 2024 – Present

Boston University Rocket Propulsion Group

Boston, MA

- Designed a converging-diverging rocket nozzle in SolidWorks, applying 1D compressible flow theory to size the throat geometry and expansion contour for an ABS-like resin print.
- Validated structural integrity via FEA under 80 psi chamber pressures, confirming a 2.11 Factor of Safety across three design reviews.
- Executed a static test-fire integrating the 3D-printed nozzle with a 1-inch NPT chamber, using existing relay and flyback diode circuitry for ignition control.
- Designing an aerodynamic fairing for an externally-mounted downcomer on the Argo rocket, evaluating supersonic profiles (tangent ogive, power law nosecones) at Mach 2.81 max-Q conditions via ANSYS CFD and OpenRocket trajectory simulation.

Project Coordinator

Jun 2026 – Aug 2026

Residential Addition & ADU Permitting Project

Los Altos, CA

- Coordinated municipal permit revisions for a residential addition and ADU remodel, using Bluebeam Revu to redline construction drawings and Excel to track compliance data.
- Evaluated schematics against California Building Code (CBC) and ANSI standards, resolving layout omissions and communicating design requirements to the owner and architect.

Academic Project

Jan 2025 – May 2025

Temperature Sensor Prototype

Boston, MA

- Developed and programmed an Arduino-based room temperature monitor in C++, featuring an LCD display, TMP36 sensor, and an alert system (LEDs, buzzer) housed in a custom ABS enclosure.
- Authored a technical report evaluating system performance, identifying key limitations (battery life, component selection, manufacturing process), and proposing improvements for potential commercialization.

SKILLS

Software: SolidWorks (FEA, HSMworks), Onshape, ANSYS CFD, OpenRocket, Python, MATLAB, C++ (Arduino), Java

Engineering: Supersonic Aerodynamics, 1D Compressible Flow, Thermodynamic Cycle Analysis, Structural Analysis, Experimental Testing

Fabrication: FDM Additive Manufacturing, CNC Machining, Manual Lathe, Milling Machine, Soldering, Electronics Assembly

Languages: English (Native), Mandarin Chinese (Conversational)

CERTIFICATIONS

A Hands-On Introduction to Engineering Simulations — edX / CornellIX, 2026

Aerodynamics Unlocked — Driver61 Education, 2025

Machine Learning Specialization — DeepLearning.AI / Stanford University, 2024